# CSIT 598 Machine Learning

Assignment 2

Due: 23:59 March 13 2026

## Objective

In this assignment, you will review key concepts regarding (classical) supervised learning and unsupervised learning. This assignment also serves as a sample exam paper for mid-term.

## 1 Multiple-Choice Questions

40 Questions; 1.5 points per question.

1. Which set is used to evaluate the effectiveness of a learned hypothesis?
   (a) Training set
   (b) Test set
   (c) Validation set
   (d) Data set
2. What is supervised learning?
   (a) Learning by self-discovery
   (b) Learning with labeled examples
   (c) Learning without guidance
   (d) Learning with reinforcement
3. Which of the following is an example of unsupervised learning?
   (a) Classifying emails as spam or not spam
   (b) Grouping objects based on similarity
   (c) Training a robot to walk
   (d) Identifying cars in images
4. In reinforcement learning, what guides the learning process?
   (a) Labeled samples
   (b) Feedback loops
   (c) Rewards
   (d) Data labels
5. Which of the following is a characteristic of inductive learning?
   (a) General principles are used to reach specific conclusions
   (b) It applies only to supervised learning
   (c) It derives general principles from specific information
   (d) It is a form of deductive reasoning
6. In decision tree induction, what is the basis for selecting test attributes?

(a) Random selection
(b) Majority voting
(c) Heuristic or statistical measure such as information gain
(d) Using only continuous attributes

7. Which of the following conditions is used to stop partitioning in decision tree construction?
   (a) When the training set becomes too large
   (b) When all samples in a node belong to the same class
   (c) When the entropy reaches zero
   (d) When all attributes are continuous
8. What does overfitting in decision trees refer to?
   (a) When the decision tree is too small to capture patterns
   (b) When the decision tree is too generic and performs poorly on training data
   (c) When the decision tree fits the training set too closely and does not generalize well to new data
   (d) When the decision tree is too balanced
9. What is a common strategy to avoid overfitting in decision trees?
   (a) Increasing the depth of the tree
   (b) Cross-validation or pruning
   (c) Decreasing the size of the dataset
   (d) Using only continuous attributes
10. How is entropy defined in machine learning?
    (a) A measure of accuracy
    (b) A measure of uncertainty associated with a random variable
    (c) A technique to prune trees
    (d) A type of classification algorithm
11. What is the main limitation of Naive Bayes classifiers?
    (a) They are too complex to implement
    (b) They cannot handle continuous data
    (c) They assume all variables are conditionally independent
    (d) They overfit the training data
12. What is the role of Bayesian belief networks?
    (a) They model dependencies among variables
    (b) They classify data based on a majority vote
    (c) They simplify decision trees
    (d) They prune decision trees
13. What is the primary goal of Support Vector Machines (SVM)?
    (a) To minimize the distance between data points
    (b) To find a decision boundary with the smallest margin
    (c) To find a decision boundary with the largest margin
    (d) To classify data without using a hyperplane
14. What are support vectors in SVM?

(a) The most distant data points from the decision boundary
(b) Data points closest to the decision boundary
(c) Data points that do not affect the decision boundary
(d) The points that lie on the hyperplane

15. What is the purpose of the kernel method in SVM?

(a) To increase the speed of computation
(b) To classify linearly separable data
(c) To map data into a higher-dimensional space
(d) To avoid overfitting

16. What is the main advantage of k-Nearest Neighbor (KNN)?

(a) It is a memory-efficient algorithm
(b) It is simple and intuitive
(c) It performs better than all other algorithms
(d) It does not require distance measures

17. How does KNN classify a new data point?

(a) By calculating the gradient between neighbors
(b) By calculating the distance to the $k$ nearest neighbors and using majority voting
(c) By dividing data into classes using hyperplanes
(d) By minimizing the loss function

18. In machine learning, what does high variance usually indicate?

(a) The model is underfitting the data
(b) The model is overfitting the data
(c) The model has a high bias
(d) The model is not sensitive to fluctuations

19. What is the goal of ensemble learning?

(a) To use a single model to make predictions
(b) To reduce the computational cost of training models
(c) To aggregate multiple models to improve predictive performance
(d) To ensure that all models are identical

20. Which of the following is a principle of ensemble learning?

(a) All models must be similar
(b) Base models should have high diversity
(c) Ensemble learning only works for classification tasks
(d) Ensemble learning does not require diversity in models

21. Which of the following is a parallel ensemble learning method?

(a) AdaBoost
(b) Gradient Boost
(c) Bagging
(d) Sequential Boosting

22. In Bagging, how are the learners trained?

(a) Sequentially, with each learner correcting the previous one

(b) Using data sampled from the original dataset with replacement
(c) On a subset of the features with no replacement
(d) By creating a single large decision tree

23. What is the main purpose of Bagging in ensemble learning?
    (a) To reduce bias
    (b) To minimize noise in the data
    (c) To reduce variance
    (d) To avoid overfitting

24. How does AdaBoost differ from Bagging?
    (a) AdaBoost trains learners in parallel, whereas Bagging trains them sequentially
    (b) AdaBoost focuses on reducing variance, while Bagging reduces bias
    (c) AdaBoost focuses on correcting difficult observations, while Bagging uses random sampling
    (d) AdaBoost uses majority voting, while Bagging uses weighted voting

25. What is the goal of AdaBoost?
    (a) To increase the variance in the ensemble
    (b) To reduce bias by focusing on hard-to-predict samples
    (c) To create identical models for comparison
    (d) To reduce the complexity of the training data

26. What is a key benefit of using Random Forest in ensemble learning?
    (a) It is highly interpretable
    (b) It reduces overfitting by using bootstrapped samples and random feature subsets
    (c) It uses gradient boosting to minimize error
    (d) It requires only a single tree to make predictions

27. What is one disadvantage of ensemble methods?
    (a) They are less accurate than individual models
    (b) They are computationally expensive and harder to interpret
    (c) They cannot handle high-dimensional data
    (d) They require a lot of hyper-parameter tuning

28. What is the main objective of regression?
    (a) To classify categorical data
    (b) To predict numeric outputs based on input features
    (c) To group similar data points into clusters
    (d) To reduce the dimensionality of data

29. Which of the following is an example of a regression task?
    (a) Predicting whether a customer will buy a product
    (b) Grouping customers based on purchasing patterns
    (c) Predicting a person's weight based on height
    (d) Identifying spam emails

30. In linear regression, what is the goal of the model?
    (a) To find a line that classifies the data
    (b) To find a line that fits the data and minimizes the sum of squared errors

(c) To group data points into clusters
(d) To create decision boundaries using hyperplanes

31. What is the role of the coefficient of determination $R^2$ in linear regression?

(a) To determine the slope of the best-fit line
(b) To measure the correlation between the input and output
(c) To measure how well the regression line fits the data
(d) To classify data into different categories

32. What is the purpose of regularization?

(a) To increase the complexity of the model
(b) To prevent overfitting by penalizing large coefficients
(c) To reduce the number of data points
(d) To simplify the regression equation

33. What is Ridge Regression?

(a) A type of regression that uses $L_1$-norm regularization
(b) A technique that uses $L_2$-norm regularization to shrink coefficients
(c) A method that eliminates features with no predictive value
(d) A method for classifying data into clusters

34. What is a common approach to choosing the regularization parameter $\lambda$?

(a) Using cross-validation
(b) Using gradient descent
(c) Setting $\lambda$ to a fixed value of 1
(d) Minimizing the number of features

35. What is the primary goal of unsupervised learning?

(a) To predict target values in the data
(b) To discover intrinsic patterns or structures in the data
(c) To classify data into predefined categories
(d) To minimize the sum of squared errors

36. What is a common measure used for clustering?

(a) Euclidean distance
(b) Mean squared error
(c) Gradient descent
(d) Loss function

37. What is a common criterion for K-means convergence?

(a) When all centroids are equidistant from each other
(b) When the centroids do not move significantly between iterations
(c) When all data points are correctly classified
(d) When the number of clusters is maximized

38. Which of the following is a technique to deal with outliers in K-means clustering?

(a) Ignore the outliers
(b) Remove data points that are far from centroids after a few iterations
(c) Increase the number of centroids

(d) Reduce the number of features

39. In hierarchical clustering, what is divisive clustering?

    (a) A method that starts with individual points and merges them into clusters
    (b) A method that starts with all data points in one cluster and splits them recursively
    (c) A method that uses centroids to find clusters
    (d) A method that eliminates outliers

40. What is the primary goal of dimensionality reduction in unsupervised learning?

    (a) To increase the number of features
    (b) To simplify the data representation by reducing the number of features
    (c) To increase the number of clusters
    (d) To perform classification on the data

# 2 Calculation Questions

1. (**Ensemble Learning, 5 points**)

   Consider $N$ independent random variables $X_1, X_2, \ldots, X_N$, each following a normal distribution with mean $\mu$ and variance $\sigma^2$, i.e., $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Define the averaged decision variable as:

$$\bar{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$$

   (a) Show that the mean of the averaged decision variable $\bar{X}$ is $\mu$.
   (b) Show that the variance of $\bar{X}$ is $\frac{\sigma^2}{N}$.
   (c) Based on derivation from (a) and (b), discuss how to reduce variance in ensemble learning (hint: consider base models as a form of random variables).

2. (**Decision Tree, 10 points**)

   Consider the following dataset for a binary classification problem where the target variable is "Play Tennis" (Yes or No), and there are two features: "Outlook" and "Temperature."

   *Note: the calculation for this is slightly harder than mid-term.*

| Outlook | Temperature | Play Tennis |
|---|---|---|
| Sunny | Hot | No |
| Sunny | Cool | Yes |
| Overcast | Cool | Yes |
| Rainy | Hot | Yes |
| Rainy | Cool | No |
| Sunny | Hot | No |
| Overcast | Hot | Yes |

   (a) Compute the entropy of the target variable "Play Tennis."
   (b) Calculate the information gain for the feature "Outlook."
   (c) Show which feature should be used as the root node based on information gain.
   (d) Create a simple decision tree based on your results.

3. (**KNN, 5 points**)

   You are given the following dataset, where each point has two features: $x_1$ and $x_2$, and a binary target class (A or B):

| Point | $x_1$ | $x_2$ | Class |
|---|---|---|---|
| $P_1$ | 1 | 1 | $A$ |
| $P_2$ | 2 | 2 | $A$ |
| $P_3$ | 3 | 3 | $B$ |
| $P_4$ | 6 | 5 | $B$ |
| $P_5$ | 7 | 6 | $B$ |

A new data point $P_{\text{new}} = (3, 4)$ arrives. Use the k-Nearest Neighbors algorithm with $k = 3$ to classify this new data point.

4. (**K-means, 10 points**)

   You are given the following dataset with two features, $x_1$ and $x_2$, representing the coordinates of points in a 2D space. Your task is to perform K-means clustering with $K = 2$ clusters:

| Point | $x_1$ | $x_2$ |
|---|---|---|
| $P_1$ | 2 | 3 |
| $P_2$ | 3 | 3 |
| $P_3$ | 6 | 8 |
| $P_4$ | 8 | 8 |

   (a) Assume initial centers are $P_2$ and $P_3$. Find the two clusters.

   (b) Recompute the centers for the two clusters, based on which find the new two clusters.

5. (**Linear Regression, 10 points**)

   You are given the following three data points: $(1, 1)$, $(2, 3)$, and $(3, 3)$. Your task is to fit a line $y = mx + b$ using the least squares method.

   (a) Calculate the slope $m$ and the intercept $b$.

   (b) Calculate the least squared error.

   *Note: round a decimal number to two decimal places.*

# 3 Bonus Question

*Note: this question is optional. Complete and correct answer to this question will add **10 points**, but the total points for assignment 2 will not exceed 100.*

Given a set of $N$ data points $(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$, where each point consists of a predictor value $x_i$ and an output value $y_i$, the goal is to find the equation of the best-fit line $y = mx + b$ that minimizes the sum of squared errors (SSE). Show how the formulas for the slope $m$ and the intercept $b$ can be derived using the method of least squares, where:

$$m = \frac{N \sum_{i=1}^{N} x_i y_i - \sum_{i=1}^{N} x_i \sum_{i=1}^{N} y_i}{N \sum_{i=1}^{N} x_i^2 - \left(\sum_{i=1}^{N} x_i\right)^2},$$

and

$$b = \frac{\sum_{i=1}^{N} y_i - m \sum_{i=1}^{N} x_i}{N}.$$

# Submission Guidelines

- Assignment should be submitted via Canvas.
- Submit a single report (either word or latex) with answers to the questions. You can choose to upload handwriting pictures in the report.